Sulabham

APPSC AEE 2026 · Module 2: Fluid Mechanics and Hydraulic Machines

Paper 2 (Civil & Mechanical common) · Post Codes 1, 2, 3, 4, 6, 7 and 8 · Study weeks 4–6 · English medium

How to use this module

1. **Understand:** read one topic a day in the order given — basic idea → why → how → analogy → traps.
2. **Connect:** look at the photo and the diagram, then read the "Connections" line to see where the idea returns in Paper 3.
3. **Remember:** learn the memory code, close the page and answer the self-recall questions aloud.
4. **Apply:** take the 5-question quiz at the end of the topic. Tap an option; the answer and the reason for every option appear only after you choose.
5. At the end, write the Notebook page from memory, then take the Module 2 test.

Week plan: Week 4 → Topics 1–4 · Week 5 → Topics 5–7 · Week 6 → Topics 8–10 + the test. No calculator is allowed in the exam, so work every number by hand.

How Fluid Mechanics was asked in APPSC AEE 2023 (Paper II, 22 Aug 2023, Shift 1)

Evidence (counted from the official paper)

Fluid Mechanics and Hydraulic Machines filled about two-thirds of Paper II (Q52 to Q150); Strength of Materials took Q1 to Q51. The table shows how many questions fell under each topic of this module. A few questions touch two topics, and two were cancelled by APPSC.

Topic	Questions	Share
Topic 1: Properties of fluids	1	
Topic 2: Pressure and its measurement	5	
Topic 3: Hydrostatic forces, buoyancy	6	
Topic 4: Fluid kinematics	20	
Topic 5: Bernoulli, momentum, flow measurement	16	
Topic 6: Flow through pipes	12	
Topic 7: Boundary layer, drag and lift	11	
Topic 8: Impact of jets and turbines	8	
Topic 9: Centrifugal pumps	10	
Topic 10: Hydro-power projects	1	

Our analysis (not a prediction)

Kinematics, Bernoulli and flow measurement, pipes and the boundary layer made up most of the Fluid Mechanics questions, so give them the most practice time. Many numericals were textbook-style (R.K. Bansal type) with numbers that simplify by hand. Statement-type and "which is INCORRECT" questions flipped one word of a true sentence. Topics with few questions in 2023 are still in the syllabus — do not skip them.

Topic 1: Properties of Fluids

Syllabus line: *Fluid properties — specific gravity, viscosity, surface tension, vapour pressure.*

Asked in APPSC AEE 2023, Paper II (verified): 1 question

- Real fluid: viscosity causes shear resistance and converts mechanical energy into heat (Q148, statement type).

1. Basic idea

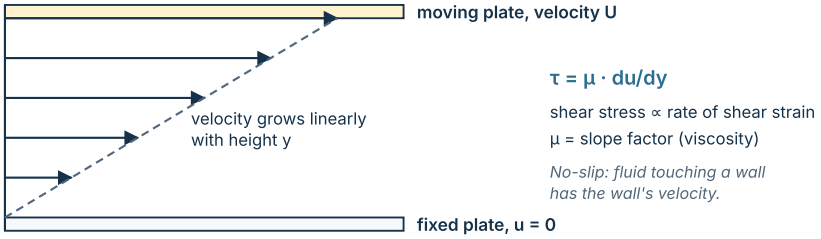
A **fluid** is anything that flows: liquids and gases. A solid resists a sideways (shear) push by deforming a little and stopping. A fluid **keeps on deforming** as long as the shear force acts — that is why water takes the shape of the bottle. Every property below tells us how a fluid behaves when we push it, squeeze it, heat it or leave it in a thin tube.

2. Why engineers need these properties

Density decides pressure (Topic 2) and buoyancy (Topic 3). Viscosity decides pipe friction losses (Topic 6) and the boundary layer (Topic 7). Vapour pressure decides cavitation in pumps and turbines (Topics 8–9). So this small topic feeds every later topic.

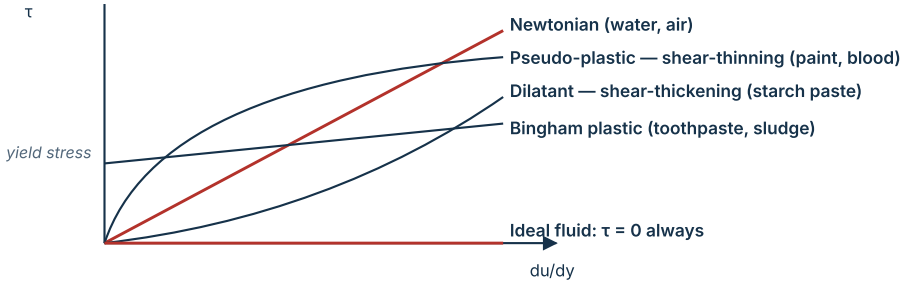
3. How each property works

Property	Meaning (simple)	Formula / unit	Remember
Mass density ρ	Mass in one cubic metre	ρ = m/V, kg/m³	Water = 1000 kg/m³ (at 4 °C)
Specific weight w	Weight of one cubic metre	w = ρg, N/m³	Water = 9810 N/m³
Specific gravity S	How many times heavier than water	$S = \rho/\rho_{water}$ (no unit)	Mercury = 13.6
Dynamic viscosity μ	Internal friction — 'stickiness'	τ = μ·du/dy, N·s/m² (Pa·s)	1 poise = 0.1 N·s/m²
Kinematic viscosity ν	Viscosity per unit density	ν = μ/ρ, m²/s	1 stoke = 1 cm²/s = 10^{-4} m²/s
Surface tension σ	Tension in the liquid 'skin'	N/m	Water–air ≈ 0.073 N/m (20 °C)
Vapour pressure p_v	Pressure at which the liquid starts boiling	N/m²	Rises with temperature
Bulk modulus K	Resistance to squeezing	K = −dp/(dV/V)	Compressibility = 1/K



How to read it: the fluid layer touching each plate moves with that plate (no-slip). In between, each layer slides over the next; the friction between layers is viscosity.

Viscosity and temperature (a favourite trap): in a **liquid**, viscosity comes from cohesion between molecules; heating loosens them, so viscosity **falls** (hot oil flows easily). In a **gas**, viscosity comes from molecules jumping between layers (momentum exchange); heating makes them jump more, so viscosity **rises**.



Only Newtonian fluids give a straight line through the origin. Bingham plastics need a minimum 'yield stress' before they flow — that is why toothpaste stays on the brush.

Essential formulas — surface tension and capillarity

- Pressure inside a **liquid droplet**: p = 4σ/d · **soap bubble** (two surfaces): p = 8σ/d · **liquid jet**: p = 2σ/d
- **Capillary rise**: h = 4σ cosθ / (ρ g d). Water–glass θ ≈ 0° → rise. Mercury–glass θ ≈ 130°–140° → cos θ negative → **fall** (depression).
- Narrower tube → higher rise (h ∝ 1/d). That is why fine pores in soil and bricks suck up water.
- **Cavitation**: when local pressure drops to the vapour pressure, the liquid boils even at room temperature; the vapour bubbles collapse violently and pit the metal (Topics 8–9).

4. Everyday analogy

Think of a pack of playing cards on a table. Push the top card sideways — every card slides a little over the one below. The "stickiness" between cards is viscosity; the bottom card stuck to the table is the no-slip condition.

5. Confusing areas and exam traps

- **Poise vs stoke:** poise is for μ (dynamic); stoke is for ν (kinematic). 1 poise = 0.1 N·s/m², not 1.
- **Ideal fluid** = no viscosity and incompressible — imaginary, used to simplify theory (Bernoulli). A **real fluid** has viscosity, so energy is lost as heat.
- **Soap bubble = 8σ/d** because it has two surfaces (inside and outside); a droplet has one.
- Specific gravity has **no unit**; specific weight has a unit (N/m³).

6. Connections

Viscosity → Reynolds number (Topic 6) → laminar or turbulent flow. Vapour pressure → NPSH and cavitation (Topic 9). Capillarity → soil mechanics (Paper 3 Civil: capillary rise in soils).

Memory code: "Liquids Loosen, Gases Get thicker" (when heated)

Liquids Loosen	Liquid viscosity decreases with temperature (cohesion weakens)
Gases Get thicker	Gas viscosity increases with temperature (more molecular momentum exchange)
4 – 8 – 2	Droplet 4σ/d · soap bubble 8σ/d · jet 2σ/d (bubble has 2 skins, so double the droplet; jet is a cylinder, so half)

Self-recall (close the notes, answer aloud, then check)

1. Write Newton's law of viscosity and name each term.
   τ = μ·du/dy — τ shear stress, μ dynamic viscosity, du/dy velocity gradient (rate of shear strain).
2. Convert 0.25 stoke into m²/s.
   $0.25 \times 10^{-4} = 2.5 \times 10^{-5}$ m²/s.
3. Why does mercury fall in a glass capillary tube?
   Its contact angle is more than 90° (≈130°–140°), so cos θ is negative and h becomes negative — adhesion with glass is weaker than cohesion of mercury.

Quick check: 5 questions

1. Consider the following statements about viscosity:
   I. Viscosity of liquids decreases as temperature increases.
   II. Viscosity of gases increases as temperature increases.
   III. 1 poise is equal to 1 N·s/m².
   IV. The unit of kinematic viscosity, the stoke, equals 1 cm²/s.

   Which of the above statements are correct?

   1. I and II only
   2. I, II and IV only
   3. II, III and IV only
   4. I, II, III and IV

   1. Leaves out IV, which is true (1 stoke = 1 cm²/s = 10^{-4} m²/s).
   2. Correct. I, II and IV are true; III is false because 1 poise = 0.1 N·s/m².
   3. Includes III, the classic unit trap.
   4. Includes the false statement III.

2. A plate 1 mm above a fixed plate moves at 0.5 m/s. A shear stress of 1 N/m² is needed to keep it moving. The dynamic viscosity of the fluid between the plates is

   1. 0.002 poise
   2. 0.02 poise
   3. 0.2 poise
   4. 2 poise

   1. This is the value in N·s/m², not in poise — the unit trap.
   2. Correct. μ = τ/(du/dy) = 1/(0.5/0.001) = 0.002 N·s/m² = 0.02 poise (1 N·s/m² = 10 poise).
   3. Ten times too large: check the conversion 1 poise = 0.1 N·s/m².
   4. Would need τ = 100 N/m².

3. Match List-I (type of fluid) with List-II (example or feature):

List-I	List-II
A. Ideal fluid	1. Toothpaste, sewage sludge
B. Newtonian fluid	2. Water, air
C. Dilatant fluid	3. Zero viscosity, incompressible
D. Bingham plastic	4. Starch paste (thickens when stirred fast)

1. A-1, B-2, C-4, D-3
2. A-3, B-4, C-2, D-1
3. A-3, B-2, C-4, D-1
4. A-4, B-3, C-2, D-1

1. Swaps the ideal fluid with the Bingham plastic.
2. Swaps Newtonian and dilatant.
3. Correct. Ideal → no viscosity; Newtonian → water and air; dilatant → starch paste (thickens when stirred fast); Bingham → toothpaste, sewage sludge (needs a yield stress).
4. Calls the ideal fluid starch paste and water dilatant — water is Newtonian.

4. Water (σ = 0.075 N/m, contact angle 0°) rises in a clean glass tube of 1 mm diameter. Taking g = 10 m/s², the capillary rise is about

1. 3 mm
2. 15 mm
3. 30 mm
4. 60 mm

1. Ten times too small: d must be in metres (0.001 m).
2. Half the answer: the formula has 4σ, not 2σ.
3. Correct. $h = 4\sigma\cos\theta/(\rho g d) = 4 \times 0.075/(1000 \times 10 \times 0.001) = 0.03$ m = 30 mm.
4. Double: uses 8σ (the soap-bubble factor).

5. **Assertion (A):** Cavitation can occur in the suction pipe of a pump even at room temperature.
Reason (R): The vapour pressure of a liquid increases with temperature.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. R is true, but it does not explain A. A happens because the local pressure falls to the vapour pressure, not because of heating.
2. Correct. Both are true; A is explained by low pressure in the suction line reaching p_v, while R is a separate fact.
3. R is true: hot water has a higher vapour pressure (it boils at 100 °C at atmospheric pressure).
4. A is true: cavitation is a low-pressure effect, not a high-temperature effect.

Topic 2: Pressure and Its Measurement

Syllabus line: *Pascal's law, pressure measurement — piezometer, U-tube and differential manometers.*

Asked in APPSC AEE 2023, Paper II (verified): 5 questions

- Mechanical gauges balance the fluid column by a spring or dead weight (Q52).
- Name the principle: pressure applied to an enclosed fluid is transmitted everywhere — Pascal's law (Q55).
- Convert a pressure of 3.924 N/cm² into head of water — 4 m (Q56).
- Pressure measured with atmospheric pressure as datum — gauge pressure (Q57); gauge pressure 3 m below a liquid surface (Q58).

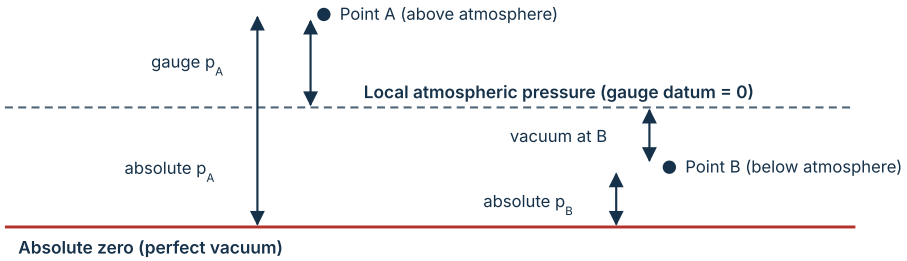
1. Basic idea

Pressure is force per unit area, p = F/A (N/m² = pascal). In a fluid at rest, pressure at a point acts **equally in all directions** (Pascal's law) and increases with depth because the fluid above has weight.

2. Why it exists

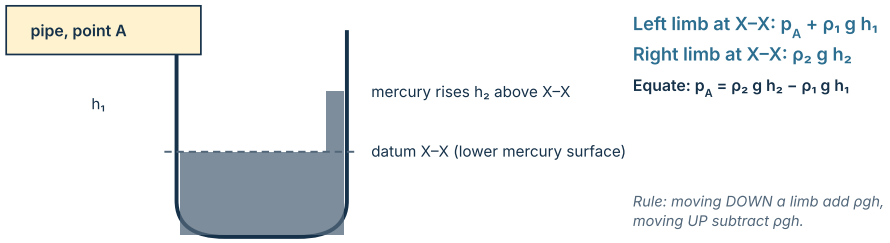
Each layer of water must carry the weight of all the water above it. Go down by h and you add the weight of a column of height h on each square metre: **p = ρ g h**. This is the **hydrostatic law**, dp/dz = −ρg (pressure falls as you go up).

3. How pressure is measured



Absolute = atmospheric + gauge. Vacuum (suction) = atmospheric − absolute. Standard atmosphere = 101.325 kN/m² = 10.3 m of water = 760 mm of mercury.

Device	How it works	Limits
Piezometer	Simple open tube; liquid rises to height h, p = ρgh	Only liquids, only moderate gauge pressure; cannot read vacuum or gas pressure
Simple U-tube manometer	Heavy liquid (mercury) in a U-tube balances the pressure	Measures high pressure and vacuum, liquids and gases
Differential manometer	U-tube connected to two points	Gives the **difference** of pressure between two points (used with venturimeter)
Inverted U-tube	Light liquid (or air) at top	Small pressure differences in liquids
Micromanometer	U-tube with an enlarged reservoir or inclined limb	Very small pressure differences
Mechanical gauges (Bourdon tube, diaphragm, dead-weight)	Pressure is balanced by a spring or a dead weight — not by a liquid column	Very high pressures (boilers, pumps); read directly



Manometer method: start at one end, add ρgh when you go down, subtract when you go up, and finish at the other end (atmosphere = 0 gauge).

Essential formulas

- p = ρ g h · head h = p/(ρg) · head of another liquid: h = p/(S × 9810) m
- $p_{abs} = p_{atm} + p_{gauge}$ · $p_{vacuum} = p_{atm} - p_{abs}$
- Differential mercury manometer with water in both limbs: head difference = x (S_{Hg}/S_W − 1) = **12.6 x**
- Hydraulic press (Pascal): F/a = W/A → W = F × (A/a)

4. Everyday analogy

Diving into a deep pond: your ears hurt more the deeper you go — more water above means more pressure. Squeezing a toothpaste tube anywhere pushes paste out at the nozzle — pressure is transmitted equally (Pascal).

5. Confusing areas and exam traps

- Gauge pressure can be negative (vacuum); absolute pressure can never be negative.
- 3.924 N/cm² = 39,240 N/m² → head = 39240/9810 = **4 m** of water. Always convert cm² to m² first (× 10^4).

- The factor 12.6 applies only when the flowing liquid is water and the manometer liquid is mercury. For oil (S = 0.8): x(13.6/0.8 − 1) = 16x.

6. Connections

Hydrostatic law → forces on dams and gates (Topic 3) → venturimeter readings (Topic 5). Soil mechanics uses the same idea for pore water pressure (Paper 3 Civil).

Memory code: "Down Add, Up Subtract" — and 10.3 / 760

Down Add	Moving downward through a liquid column: add ρgh
Up Subtract	Moving upward: subtract ρgh
10.3 / 760	1 atmosphere = 10.3 m of water = 760 mm of mercury (= 101.325 kN/m²)
12.6	Mercury under water: 13.6 − 1

Self-recall (close the notes, answer aloud, then check)

1. State Pascal's law and one practical use.
   Pressure at a point in a fluid at rest is equal in all directions / transmitted equally throughout an enclosed fluid. Uses: hydraulic jack, press, brakes.
2. A vacuum gauge reads 30 kN/m² where the atmosphere is 100 kN/m². What is the absolute pressure?
   100 − 30 = 70 kN/m² (absolute).
3. Why can a piezometer not measure the pressure of a gas?
   The gas would escape through the open tube; a liquid column is needed to balance the pressure, so a U-tube with a heavier liquid is used.

Quick check: 5 questions

1. A pressure gauge on a pipe carrying oil (specific gravity 0.8) reads 98.1 kN/m². The equivalent head of oil is

   1. 8 m
   2. 10 m
   3. 12.5 m
   4. 80 m

   1. Multiplies by 0.8 instead of dividing.
   2. This is the head of water, not oil.
   3. Correct. h = p/(ρg) = 98,100/(800 × 9.81) = 12.5 m of oil. A lighter liquid needs a taller column.
   4. Unit slip — forgets ρg completely.

2. The barometer reads 750 mm of mercury and a vacuum gauge on a condenser reads 600 mm of mercury. The absolute pressure in the condenser is about

   1. 20 kN/m²
   2. 80 kN/m²
   3. 100 kN/m²
   4. 180 kN/m²

   1. Correct. p_{abs} = 750 − 600 = 150 mm Hg = 0.150 × 13,600 × 9.81 ≈ 20 kN/m².
   2. This is 600 mm Hg (the vacuum itself), not the absolute pressure.
   3. Roughly the atmospheric pressure — ignores the vacuum.
   4. Adds the vacuum to the atmosphere — vacuum must be subtracted.

3. Consider the following statements about pressure-measuring devices:
   I. A piezometer cannot measure negative (vacuum) pressure.
   II. A piezometer is unsuitable for high pressures of liquids.
   III. A piezometer is suitable for measuring gas pressure.
   IV. A differential manometer measures the difference of pressure between two points.
   Which of the above statements are correct?

   1. I and II only
   2. II, III and IV only
   3. I, II, III and IV
   4. I, II and IV only

   1. Leaves out IV, which is true.
   2. Includes III — a gas escapes through an open piezometer tube.
   3. Includes the false statement III.
   4. Correct. I, II and IV are true; III is false.

4. A differential mercury manometer is connected to two points of a horizontal pipe carrying water. The mercury levels differ by 15 cm. The difference of pressure head between the two points is

1. 0.15 m of water
2. 2.04 m of water
3. 1.74 m of water
4. 1.89 m of water

1. This is the mercury reading, not the water head.
2. Uses 13.6 instead of (13.6 − 1).
3. Uses an arbitrary factor — check: 0.15 × 11.6 = 1.74.
4. Correct. $h = x(S_{Hg}/S_w - 1) = 0.15 \times 12.6 = 1.89$ m of water (the same idea was asked in 2023 Q69 with a venturimeter).

5. Match List-I with List-II:

List-I	List-II
A. Pascal's law	1. Very small pressure differences
B. Hydrostatic law	2. Hydraulic jack
C. Bourdon tube gauge	3. $dp/dz = -\rho g$
D. Micromanometer	4. Mechanical (spring-type) gauge

1. A-2, B-3, C-4, D-1
2. A-3, B-2, C-4, D-1
3. A-2, B-3, C-1, D-4
4. A-3, B-2, C-1, D-4

1. Correct. Pascal's law → hydraulic jack; hydrostatic law → $dp/dz = -\rho g$; Bourdon tube → spring-type mechanical gauge; micromanometer → very small pressure differences.
2. Swaps Pascal's law and the hydrostatic law.
3. Swaps the Bourdon gauge and the micromanometer.
4. Swaps both pairs.

Topic 3: Hydrostatic Forces, Buoyancy and Floating Bodies

Syllabus line: *Hydrostatic forces on plane and curved surfaces; buoyancy, floating bodies and stability.*

Asked in APPSC AEE 2023, Paper II (verified): 6 questions (+1 cancelled)

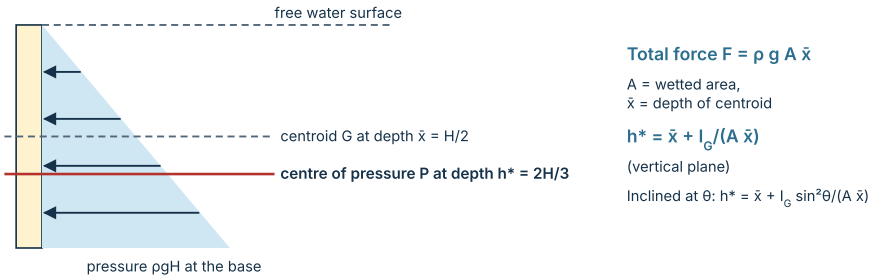
- Centre of pressure of an inclined rectangular plane surface (Q54).
- Total pressure on a vertical rectangular plate whose top edge is 1 m below the surface (Q60).
- Formula for depth of centre of pressure, $h^* = I_G/(A\,\bar{x}) + \bar{x}$ (Q61).
- Weight of a floating body from its depth of immersion (Q62); meaning of metacentre (Q63); time period of rolling (Q64).

1. Basic idea

Water pushes on every wetted surface — a dam, a sluice gate, the side of a tank. Because pressure grows with depth, the push is **larger at the bottom**. We need two answers: **how big** is the total force, and **where** it acts (the **centre of pressure**).

2. Why the centre of pressure is below the centroid

Pressure is small near the top and large near the bottom (a triangle of pressure). The resultant of a triangle acts at its lower third, so the force acts **below** the centroid of the plate. Only for a horizontal plate (uniform pressure) do the two points coincide.



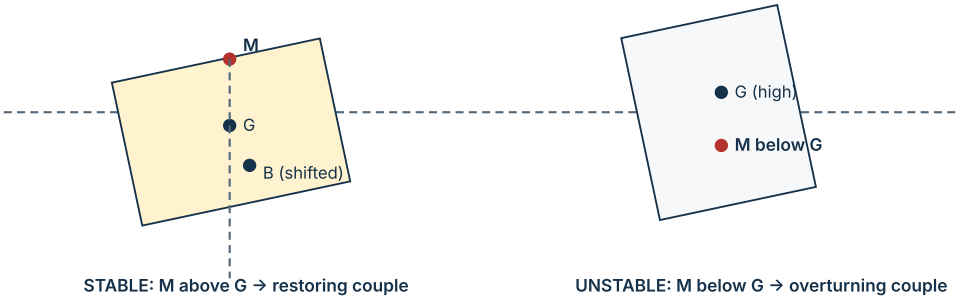
For a vertical rectangle with its top edge at the surface, the pressure diagram is a triangle, so the force acts at 2/3 of the depth.

3. Curved surfaces (gates, pipes, domes)

- **Horizontal component** F_H = force on the **vertical projection** of the curved surface (use $F = \rho g A \bar{x}$ on that projection).
- **Vertical component** F_V = **weight of the liquid** actually or imaginarily lying **above** the curved surface up to the free surface.
- Resultant $F = \sqrt{(F_H^2 + F_V^2)}$, inclined at $\tan^{-1}(F_V/F_H)$.

4. Buoyancy and floating bodies

Archimedes' principle: a body in a fluid is pushed up by a force equal to the **weight of fluid displaced**. This upward force acts at the **centre of buoyancy B** — the centroid of the displaced volume. A floating body: weight = buoyancy, so **depth of immersion = S_{body}/S_{liquid} × depth**.



When a floating body tilts, B moves to the new centroid of displaced water. The vertical line through the new B cuts the centre line at the metacentre M. GM = metacentric height.

Essential formulas — stability

- Metacentric height **$GM = I/V - BG$** (I = second moment of the water-line plane about the tilt axis, V = volume displaced)
- **Floating body:** stable if M is above G (GM > 0), neutral if M coincides with G, unstable if M is below G.
- **Fully submerged body** (submarine, balloon): stable if B is above G.
- Time period of rolling **$T = 2\pi \sqrt{(k^2 / (g \cdot GM))}$** (k = radius of gyration). Large GM → quick, jerky rolling; small GM → slow, comfortable rolling but less safe.

5. Everyday analogy

A loaded country boat with passengers standing up feels tippy — G has gone up and GM has shrunk. Ask them to sit on the floor and the boat steadies: G comes down and GM increases.

6. Confusing areas and exam traps

- Total force **$F = \rho g A \bar{x}$ does not depend on the inclination**; the centre of pressure does ($\sin^2\theta$ term).
- As a plate goes deeper, $I_G/(A\,\bar{x})$ gets smaller, so the centre of pressure moves **closer** to the centroid.
- Metacentre M is a point; metacentric height is the distance GM. The 2023 paper asked to tell them apart (Q63).

7. Connections

Dam and gate design (Paper 3 Civil — gravity dams: water pressure is the main force); retaining walls (earth pressure uses the same triangle idea); ship design.

Memory code: "2/3 down, M above G is Good"

2/3 down	Vertical wall, water from the top: force acts at 2H/3 depth
M above G is Good	Floating body stable when the metacentre is above the centre of gravity
Submerged: B above G	For a fully submerged body, buoyancy centre must be above G

Self-recall (close the notes, answer aloud, then check)

1. Write the formula for depth of centre of pressure of an inclined plane surface.
   $h^* = \bar{x} + I_G \sin^2\theta/(A\,\bar{x})$.
2. A vertical gate 2 m wide and 3 m deep has its top edge at the water surface. Find the force and where it acts.
   $F = 9810 \times 6 \times 1.5 = 88.29$ kN; $h^* = 1.5 + (2\times27/12)/(6\times1.5) = 1.5 + 0.5 = 2$ m (= 2H/3).
3. What is the vertical force on a curved surface equal to?
   The weight of liquid (real or imaginary) lying vertically above the surface up to the free surface.

Quick check: 5 questions

1. A vertical rectangular gate 2 m wide and 3 m deep has its upper edge at the water surface. The total pressure force and the depth of the centre of pressure are

   1. 58.9 kN and 2.0 m
   2. 88.3 kN and 1.5 m
   3. 176.6 kN and 2.25 m
   4. 88.3 kN and 2.0 m

   1. Uses $\bar{x}$ = 1 m — the centroid of a 3 m deep plate is at 1.5 m.
   2. Puts the force at the centroid — the centre of pressure is always lower.
   3. Uses the full depth (3 m) as $\bar{x}$.
   4. Correct. $F = \rho g A\bar{x} = 9810 \times 6 \times 1.5 = 88.29$ kN; $h^* = \bar{x} + I_G/(A\bar{x}) = 1.5 + 4.5/9 = 2.0$ m (= 2H/3).

2. Consider the following statements about a plane surface immersed in a liquid:

   I. The centre of pressure lies below the centroid unless the surface is horizontal.
   II. As the depth of immersion increases, the centre of pressure moves closer to the centroid.
   III. The total pressure force depends on the angle of inclination for the same depth of centroid.
   IV. For a horizontal surface, the centre of pressure coincides with the centroid.

   Which of the above statements are correct?

   1. I and II only
   2. I, II and IV only
   3. I, II and III only
   4. I, II, III and IV

   1. Leaves out IV, which is true.
   2. Correct. I, II and IV are true. III is false: $F = \rho g A\bar{x}$ has no θ term.
   3. Includes III and leaves out IV.
   4. Includes the false statement III.

3. A block of wood of specific gravity 0.7 and 1 m depth floats in an oil of specific gravity 0.875. Its depth of immersion is

   1. 0.6 m
   2. 0.7 m
   3. 0.8 m
   4. 0.875 m

   1. No combination of the given data gives 0.6.
   2. This would be the immersion in water, not in oil.
   3. Correct. Weight = buoyancy → 0.7 × 1 = 0.875 × h → h = 0.8 m.
   4. Uses only the oil's specific gravity.

4. **Assertion (A):** A passenger ship is given only a small metacentric height.
Reason (R): The time period of rolling is inversely proportional to the square root of the metacentric height.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. Large GM → small T → quick, jerky rolling that is uncomfortable. So passenger ships keep GM small (but positive). R explains A.
2. R is the direct reason, not an unrelated fact.
3. R is true: $T = 2\pi\sqrt{(k^2/(g \cdot GM))}$.
4. A is true: comfort needs slow rolling, i.e. a small but positive GM.

5. **Statement 1:** The vertical component of the hydrostatic force on a curved surface equals the weight of liquid lying vertically above it up to the free surface.
Statement 2: The horizontal component equals the force on the vertical projection of the curved surface.

1. Only Statement 1 is correct
2. Only Statement 2 is correct
3. Both statements are correct
4. Neither statement is correct

1. Statement 2 is also correct.
2. Statement 1 is also correct.
3. Correct. Both are the standard rules for curved surfaces (gates, domes, pipes).
4. Both are textbook results.

Topic 4: Fluid Kinematics — Types of Flow, Continuity, Stream Function and Flow Net

Syllabus line: *Kinematics — types of flow, velocity and acceleration, continuity equation, velocity potential, stream function, flow net.*

Asked in APPSC AEE 2023, Paper II (verified): 20 questions — the largest block in Fluid Mechanics

- Continuity: velocity after the diameter changes (Q98, Q100); compressible flow with area and density change (Q99); two pipes joining into one (Q101).
- Velocity components from a stream function $\psi = 5x - 6y$ (Q103); condition for irrotational flow (Q97); possible irrotational ψ (Q111).
- Rotation, vorticity (= 2 × rotation), shear strain rate (Q106, Q112, Q113, Q117).
- Local and convective acceleration (Q114, Q116); finding the third velocity component (Q115).
- Flow net — uses and statements (Q104, Q105); type of flow when a tank drains (Q107); which parameters exist in both rotational and irrotational flow (Q108).

1. Basic idea

Kinematics describes **how** fluid moves — velocity and acceleration — without asking what forces cause the motion (that is dynamics, Topic 5). We describe the velocity at every point: $V = u\,i + v\,j + w\,k$.

2. Types of flow (learn in pairs)

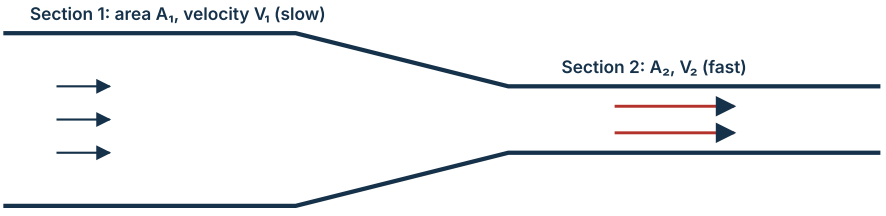
Pair	Test	Example
Steady / Unsteady	Does velocity at a fixed point change with time? $\partial V/\partial t = 0$ → steady	Constant-head tank outflow is steady; a draining tank is unsteady
Uniform / Non-uniform	Does velocity change from point to point along the flow at an instant? $\partial V/\partial s = 0$ → uniform	Long straight pipe of constant diameter is uniform; a tapering pipe is non-uniform
Laminar / Turbulent	Smooth layers vs random mixing (Reynolds number, Topic 6)	Honey pouring vs river in flood
Compressible / Incompressible	Does density change? ρ = constant → incompressible	Water (incompressible); air at high speed (compressible)
Rotational / Irrotational	Do fluid particles spin about their own axis? $\omega = 0$ → irrotational	Flow far from walls ≈ irrotational; flow in a boundary layer is rotational

Pair	Test	Example
1-D / 2-D / 3-D	Number of space coordinates the velocity depends on	Pipe flow is treated as 1-D

2023 example (Q107): water drains from a tank through a small pipe of uniform diameter. Along the pipe the velocity is the same (uniform), but it falls as the tank level drops (unsteady) → **unsteady and uniform**.

3. Continuity — mass is never lost

Whatever mass enters a section per second must leave the next section: $\rho_1 A_1 V_1 = \rho_2 A_2 V_2$. For a liquid (ρ constant): $A_1V_1 = A_2V_2 = Q$. Since $A \propto d^2$, $V \propto 1/d^2$: halve the diameter → four times the velocity.



The same discharge squeezes through a smaller area, so it must move faster — the same reason a garden hose jet speeds up when you press the nozzle.

Essential formulas — continuity and acceleration

- 3-D continuity (incompressible): **$\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$** — use it to find a missing velocity component (2023 Q115).
- Acceleration = **local** ($\partial u/\partial t$, change with time) + **convective** ($u\partial u/\partial x + v\partial u/\partial y + w\partial u/\partial z$, change with position).
- 1-D steady flow along a streamline: $a = V \cdot dV/ds$. Example (2023 Q116): V rises uniformly from 2 to 5 m/s over 1 m → at B, $a = 5 \times 3/1 = 15$ m/s².
- Two pipes (each d, velocity v) joining one pipe D with velocity v/3: $2(\pi d^2/4)v = (\pi D^2/4)(v/3) \rightarrow D^2 = 6d^2$ (2023 Q101).

4. Rotation, vorticity and strain (the "spin" of a particle)

Advanced / exam extension — rotation formulas (2-D, x–y plane)

- Rotation about z: **$\omega_z = \frac{1}{2}(\partial v/\partial x - \partial u/\partial y)$**
- Vorticity = **2 × rotation** = $\partial v/\partial x - \partial u/\partial y$
- Irrotational flow: $\omega = 0 \rightarrow$ **$\partial v/\partial x = \partial u/\partial y$**
- Shear strain rate (Bansal): $\frac{1}{2}(\partial v/\partial x + \partial u/\partial y)$. Example (2023 Q106): $u = 8x^3$, $v = -10x^2y \rightarrow \frac{1}{2}(-20xy + 0) = -10xy$.

5. Stream function ψ and velocity potential φ

	Stream function ψ	Velocity potential φ
Velocity components (Bansal sign convention, used in the 2023 key)	$u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$	$u = -\partial\varphi/\partial x$, $v = -\partial\varphi/\partial y$
Exists when	Flow satisfies continuity (2-D) — **rotational or irrotational**	Flow is **irrotational** only
Its lines	ψ = constant are **streamlines**; difference of ψ between two streamlines = discharge per unit width	φ = constant are **equipotential lines**
Laplace equation $\nabla^2 = 0$	Satisfied if the flow is irrotational	Satisfied if the flow satisfies continuity

Worked example (2023 Q103): $\psi = 5x - 6y \rightarrow u = -\partial\psi/\partial y = -(-6) = 6$ units/s; $v = \partial\psi/\partial x = 5$ units/s. *Note: some books use the opposite sign; the 2023 official key followed the convention above, so learn this one.*

Flow net: a grid of streamlines and equipotential lines that cut each other at **right angles**. It needs irrotational flow and fixed boundaries. Uses: **seepage under dams and weirs**, uplift pressure, design of hydraulic structures, exit gradient. It changes if the boundary changes and cannot be drawn where the flow separates (2023 Q104, Q105).

6. Streamline, pathline, streakline

- **Streamline:** an imaginary line whose tangent gives the velocity direction at that instant.
- **Pathline:** the actual path traced by one particle over time.
- **Streakline:** the line joining all particles that passed through one fixed point (dye or smoke injected at a point).
- In **steady** flow all three coincide.

7. Confusing areas and exam traps

- Vorticity is **twice** the rotation — options often give "equal to rotation" (2023 Q117).
- ψ exists in both rotational and irrotational flow; φ exists only in irrotational flow. Vorticity is defined for both (2023 Q108: answer "vorticity and stream function").
- For a compressible fluid use ρAV, not AV (2023 Q99: area falls 40 %, density falls 15 % → $V_2/V_1 = 1/(0.6 \times 0.85) = 1.96$).

8. Connections

Flow net → seepage and uplift under weirs (Paper 3 Civil irrigation: Bligh and Khosla theories); continuity → every pipe and channel problem; acceleration → Euler's equation (Topic 5).

Memory code: “Stream Survives Spin, Potential needs Peace”

Stream Survives Spin	ψ exists even when the flow is rotational (only continuity needed)
Potential needs Peace	φ exists only when the flow is irrotational (no spin)
Vorticity = 2ω	Vorticity is double the rotation
Right-angle net	Streamlines ⊥ equipotential lines in a flow net

Self-recall (close the notes, answer aloud, then check)

1. Write the condition for a 2-D flow to be irrotational.
   $\partial v/\partial x - \partial u/\partial y = 0$, i.e. $\partial v/\partial x = \partial u/\partial y$.
2. If the diameter of a pipe reduces to one-quarter, what happens to the velocity?
   $V \propto 1/d^2$ → velocity becomes 16 times (2023 Q100).
3. Name two uses of a flow net.
   Seepage quantity under dams/weirs, uplift pressure and exit gradient for designing hydraulic structures.

Quick check: 5 questions

1. Water flows through a pipe that reduces from 20 cm diameter to 10 cm diameter. If the velocity in the 20 cm pipe is 2 m/s, the velocity in the 10 cm pipe is

   1. 4 m/s
   2. 8 m/s
   3. 16 m/s
   4. 0.5 m/s

   1. Uses the diameter ratio, not the area ratio.
   2. Correct. $V_2 = V_1(d_1/d_2)^2 = 2 \times 2^2 = 8$ m/s.
   3. Squares the ratio twice.
   4. Inverts the ratio — the smaller pipe must carry faster flow.

2. For a 2-D flow with u = 2x and v = −2y, consider the following statements:
   I. The flow satisfies the continuity equation.
   II. The flow is irrotational.
   III. A velocity potential function exists.
   IV. A stream function exists.
   Which of the above statements are correct?

   1. I and II only
   2. I and III only
   3. III and IV only
   4. I, II, III and IV

   1. Leaves out III and IV, which also hold.
   2. Leaves out II and IV.
   3. Leaves out I and II.
   4. Correct. $\partial u/\partial x + \partial v/\partial y = 2 - 2 = 0$ (continuity, so ψ exists); $\partial v/\partial x - \partial u/\partial y = 0 - 0 = 0$ (irrotational, so φ exists).

3. For the 2-D flow u = 3y, v = 2x, the rotation component ω_z is

1. −0.5
2. 0.5
3. 2.5
4. −1.0

1. Correct. $\omega_z = \frac{1}{2}(\partial v/\partial x - \partial u/\partial y) = \frac{1}{2}(2 - 3) = -0.5$.
2. Sign error: it is ∂v/∂x − ∂u/∂y, not the reverse.
3. Adds the derivatives — that is the shear strain rate.
4. This is the vorticity (2ω), not the rotation.

4. Match List-I (type of flow) with List-II (condition):

List-I	List-II
A. Steady flow	1. ρ = constant
B. Uniform flow	2. ω = 0
C. Irrotational flow	3. ∂V/∂t = 0 at a point
D. Incompressible flow	4. ∂V/∂s = 0 at an instant

1. A-4, B-3, C-2, D-1
2. A-3, B-4, C-1, D-2
3. A-3, B-4, C-2, D-1
4. A-4, B-3, C-1, D-2

1. Swaps steady and uniform.
2. Swaps irrotational and incompressible.
3. Correct. Steady → no change with time at a point; uniform → no change with distance at an instant; irrotational → zero rotation; incompressible → constant density.
4. Swaps both pairs.

5. In a steady one-dimensional flow, the velocity increases uniformly from 2 m/s at A to 6 m/s at B, which is 2 m downstream. The acceleration of the fluid at B is

1. 2 m/s²
2. 4 m/s²
3. 12 m/s²
4. 24 m/s²

1. This is only dV/ds; it must be multiplied by V.
2. Uses the average velocity incorrectly.
3. Correct. Convective acceleration a = V(dV/ds) = 6 × (4/2) = 12 m/s². Local acceleration is zero (steady).
4. Doubles the gradient.

Topic 5: Fluid Dynamics — Bernoulli, Momentum and Flow Measurement

Syllabus line: *Euler's and Bernoulli's equations and applications; momentum equation; flow measurement — venturimeter, orificemeter, rotameter, pitot tube, hot-wire anemometer.*

Asked in APPSC AEE 2023, Paper II (verified): 16 questions

- Total head of water from pressure, velocity and datum (Q53, Q85); piezometric head = elevation + pressure head (Q70).
- Assumptions of Bernoulli and Euler equations — which is INCORRECT (Q65, Q73); HGL and TEL definitions and statements (Q71, Q86, Q94, Q95).
- Venturimeter with a differential mercury manometer (Q69); negative pressure at the throat (Q72); orifice meter statements (Q67).
- Pitot tube — measures point velocity (Q68, Q83); velocity from stagnation and static heads (Q74); moment-of-momentum statements (Q75).

1. Basic idea

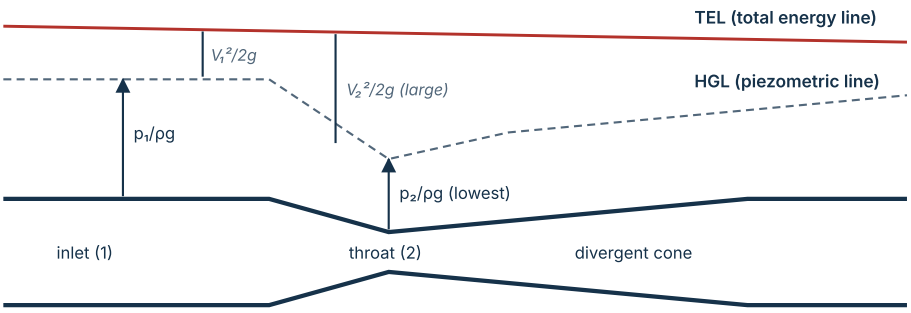
Flowing water carries three kinds of energy per unit weight (each measured in metres, called **head**): **pressure head** p/ρg, **velocity (kinetic) head** $V^2/2g$ and **datum (potential) head** z. **Bernoulli's equation** says that for an ideal fluid their sum stays constant along a streamline:

$$p/\rho g + V^2/2g + z = H = \text{constant}$$

2. Why it is true

It is the law of conservation of energy. Start from **Euler's equation** (Newton's second law for a fluid particle moving along a streamline), $dp/\rho + g\,dz + V\,dV = 0$, and integrate for an incompressible fluid. Assumptions: **ideal (non-viscous), steady, incompressible, along a streamline, irrotational**. So "the flow is rotational" or "the flow is unsteady" is the INCORRECT assumption (2023 Q65, Q73).

3. How it is applied: HGL and TEL



TEL sits $V^2/2g$ above the HGL. At the narrow throat the velocity head is large, so the pressure head (HGL) dips — it can even fall below atmospheric (and to vapour pressure, causing cavitation; 2023 Q72). The TEL slopes gently down because of friction.

- **HGL** (hydraulic gradient line) joins the piezometric heads $p/\rho g + z$. It can rise or fall (it rises after an enlargement as velocity head turns back into pressure).
- **TEL** (total energy line) joins $p/\rho g + V^2/2g + z$. It always **falls** in the direction of flow, unless a pump adds energy. TEL is always above HGL by $V^2/2g$.
- A sudden change of slope of the HGL in a straight pipe of constant diameter shows a **leakage** (discharge — and so friction loss per metre — changes there; 2023 Q84).

Essential formulas — flow measurement

- **Venturimeter**: $Q = C_d \cdot A_1A_2\sqrt{(2gh)} / \sqrt{(A_1^2 - A_2^2)}$; $C_d \approx 0.95$–0.99 (gradual cones, little loss). Mercury manometer reading x → h = 12.6 x (water).
- **Orifice meter**: same formula with A_o (orifice) and $C_d \approx 0.6$–0.65; orifice diameter usually 0.5 × pipe diameter (0.4–0.8). Cheap and compact, but larger head loss.
- **Pitot tube** measures **velocity at a point**: $V = C_v\sqrt{(2g\Delta h)}$, Δh = stagnation head − static head.
- **Rotameter**: a float rises in a tapered glass tube until the flow area balances its weight — variable-area meter, read directly.
- **Hot-wire anemometer**: a heated fine wire is cooled by the flow; the cooling gives velocity and its rapid fluctuations (turbulence in gases).
- **Momentum equation**: $\Sigma F = \rho Q(V_2 - V_1)$ (force on bends, nozzles, jets). **Moment of momentum**: $T = \rho Q(V_2r_2 - V_1r_1)$ — turbines, pumps, lawn sprinklers (2023 Q75).
- Correction factors for laminar pipe flow: kinetic energy $\alpha = 2$, momentum $\beta = 4/3$.

Advanced / exam extension — vortex flow (asked in 2023, Q118)

- **Forced vortex** (liquid rotated by an outside torque, e.g. a stirred or spinning tank): $v = \omega r$; free surface is a paraboloid.
- Depth of the paraboloid $\mathbf{z = \omega^2R^2/2g}$. The volume of a paraboloid is half that of the enclosing cylinder, so the liquid at the centre falls by z/2 and rises at the wall by z/2.
- 2023 Q118: tank of 20 cm diameter at 300 rpm → $\omega = 2\pi \times 300/60 = 31.4$ rad/s; $z = 31.4^2 \times 0.1^2/(2 \times 9.81) = 0.503$ m ≈ 50.3 cm.
- **Free vortex** (no outside torque, e.g. water draining through a sink hole, whirlpool): $v \times r$ = constant; Bernoulli's equation applies across streamlines.

4. Everyday analogy

Money in three pockets: pressure, speed and height. You can move money between pockets (the throat moves pressure into speed), but friction is a small fee that is never returned — so the total (TEL) slowly falls.

5. Confusing areas and exam traps

- Pitot tube → **point velocity**. Venturimeter and orifice meter → **discharge**. Piezometer → pressure.
- Throat velocity is **maximum** and pressure **minimum**. (2023 Q72 S1 wrongly said the velocity is minimum, so S1 was false.)
- Venturimeter C_d (≈0.98) > orifice meter C_d (≈0.62) because the gradual divergent cone avoids eddies.

6. Connections

Bernoulli + continuity solve almost every pipe problem (Topic 6). Momentum → force of jets on vanes and turbines (Topic 8). Same energy idea in open channels — specific energy (Paper 3 Civil hydraulics).

Memory code: "S-I-N-S" for Bernoulli's assumptions

S	Steady flow
I	Incompressible (and irrotational) flow
N	Non-viscous — ideal fluid
S	Along a Streamline
PV-DR	Pitot → Velocity (point); venturi/orifice → Discharge (Rate)

Self-recall (close the notes, answer aloud, then check)

1. Write the expression for total head and name each term.
   $H = p/\rho g$ (pressure head) + $V^2/2g$ (velocity head) + z (datum head).
2. Water at 2 m/s, pressure 2×10^5 N/m², 4 m above datum (g = 10). Total head?
   20 + 0.2 + 4 = 24.2 m (2023 Q85).
3. Why does a venturimeter have a long divergent cone?
   To slow the flow gradually, so that the flow does not separate and the loss of energy is small.

Quick check: 5 questions

1. Water flows in a pipe at 4 m/s under a pressure of 200 kN/m² at a section 5 m above the datum. Taking g = 10 m/s², the total head at the section is

   1. 25.8 m
   2. 25.0 m
   3. 20.8 m
   4. 26.6 m

   1. Correct. $H = p/\rho g + V^2/2g + z = 20 + 0.8 + 5 = 25.8$ m.
   2. Leaves out the velocity head.
   3. Leaves out the datum head.
   4. Uses V^2/g instead of $V^2/2g$.

2. A pitot-static tube in a pipe shows a difference of 0.45 m of water between the stagnation and static heads. If the coefficient of the tube is 0.98 (g = 10 m/s²), the velocity is

   1. 3.0 m/s
   2. 2.12 m/s
   3. 4.34 m/s
   4. 2.94 m/s

   1. Ignores the coefficient 0.98.
   2. Uses $\sqrt{(g\Delta h)}$ — forgets the 2.
   3. Uses a head difference of 1 m.
   4. Correct. $V = C_v\sqrt{(2g\Delta h)} = 0.98 \times \sqrt{(2 \times 10 \times 0.45)} = 0.98 \times 3 = 2.94$ m/s.

3. Consider the following statements:

I. The total energy line always lies above the hydraulic gradient line by the velocity head.

II. The total energy line can rise in the direction of flow without any pump.

III. The hydraulic gradient line can rise in the direction of flow at a sudden enlargement.

IV. At the throat of a venturimeter, the pressure may fall below atmospheric.

Which of the above statements are correct?

1. I and II only
2. I, II and III only
3. I, III and IV only
4. I, II, III and IV

1. Leaves out III and IV.
2. Includes II — energy cannot rise unless it is added (pump).
3. Correct. I, III and IV are true; II is false.
4. Includes the false statement II.

4. Match List-I (device) with List-II (key feature):

List-I	List-II
A. Venturimeter	1. Velocity at a point
B. Orifice meter	2. C_d about 0.62
C. Pitot tube	3. Rapid velocity fluctuations in a gas
D. Hot-wire anemometer	4. C_d about 0.95–0.99

1. A-2, B-4, C-1, D-3
2. A-4, B-2, C-1, D-3
3. A-4, B-2, C-3, D-1
4. A-2, B-4, C-3, D-1

1. Swaps venturimeter and orifice meter.
2. Correct. Venturimeter → C_d 0.95–0.99; orifice meter → C_d about 0.62; pitot tube → velocity at a point; hot-wire anemometer → rapid velocity fluctuations in a gas.
3. Swaps pitot tube and hot-wire anemometer.
4. Swaps both pairs.

5. Which of the following is NOT an assumption of Bernoulli's equation in its basic form?

1. The fluid is ideal (non-viscous).
2. The flow is steady.
3. The flow is incompressible.
4. The flow is rotational.

1. This IS an assumption.
2. This IS an assumption.
3. This IS an assumption.
4. Correct (this is the answer). Bernoulli's equation between any two points needs irrotational flow; rotational is the wrong assumption (asked as 'INCORRECT' in 2023 Q65).

Topic 6: Flow Through Pipes — Losses, Pipe Networks and Water Hammer

Syllabus line: *Pipe flow — Reynolds experiment, major and minor losses, pipes in series and parallel, TEL and HGL, water hammer.*

Asked in APPSC AEE 2023, Paper II (verified): 12 questions

- Equivalent length for minor losses $5V^2/2g$ with f = 0.04 in an 8 cm pipe — 10 m (Q87); discharge in a parallel branch (Q88).
- Head loss by Chezy's formula (Q89); loss at a sudden enlargement (Q90); ratio of loss to velocity head when the diameter doubles — 9/16 (Q91); loss due to an obstruction (Q92).
- Rise of pressure when a valve closes in 25 s (Q93); condition for maximum efficiency of power transmission — h_f = H/3 (Q96, Q120); power at the outlet of a pipe (Q122).
- Critical velocity for an oil of viscosity 0.25 stoke in a 10 cm pipe — 0.5 m/s (Q109); leakage shown by a change in HGL slope (Q84).

1. Basic idea

Real water rubs against the pipe wall and against itself, so it **loses head** as it flows. Losses along the length are **major losses** (friction); losses at fittings — bends, valves, enlargements, entrances — are **minor losses**.

2. Laminar or turbulent? — Reynolds' experiment (1883)

Osborne Reynolds injected dye into water flowing in a glass tube. At low speed the dye stayed as a **straight thread** (laminar). As the speed rose, the thread wavered and then **mixed completely** (turbulent). What decides this is the **Reynolds number** — the ratio of inertia force to viscous force:

$Re = \rho VD/\mu = VD/\nu$ $Re < 2000$ laminar · 2000–4000 transition · $Re > 4000$ turbulent (pipes)



In laminar flow the profile is a parabola and the centre-line velocity is twice the average. In turbulent flow mixing evens out the velocity, so the profile is flat in the middle and steep near the wall.

Essential formulas — major (friction) loss

- **Darcy–Weisbach**: $h_f = f\,L\,V^2 / (2\,g\,D)$, f = Darcy friction factor. (Some Indian books write $h_f = 4\,f'\,L\,V^2/(2gD)$ with $f' = f/4$ — check which f the question gives.)
- **Laminar flow**: $f = 64/Re$; Hagen–Poiseuille $h_f = 32\,\mu\,V\,L / (\rho\,g\,D^2)$ — loss $\propto V$.
- **Chezy's formula**: $V = C\sqrt{(m\,i)}$, hydraulic mean depth $m = A/P = D/4$ for a full pipe, $i = h_f/L$.
- **Critical velocity**: $V = Re_c\,\nu / D$ ($Re_c = 2000$). 0.25 stoke in 10 cm pipe → $2000 \times 0.25\times10^{-4} / 0.1 = 0.5$ m/s.

Essential formulas — minor losses

- **Sudden enlargement**: $h_e = (V_1 - V_2)^2 / 2g$. Diameter doubles → $V_2 = V_1/4$ → loss $= (3/4)^2\,V_1^2/2g =$ **9/16** of the inlet velocity head.
- **Sudden contraction**: $h_c = (1/C_c - 1)^2\,V_2^2/2g \approx 0.5\,V_2^2/2g$ (when C_c is not given, use 0.5).
- **Entrance** $= 0.5\,V^2/2g$ · **exit** $= V^2/2g$ · **bend / fitting** $= k\,V^2/2g$ · **obstruction** $= [A/(C_c(A - a)) - 1]^2\,V^2/2g$
- **Equivalent length** of a minor loss $kV^2/2g$: $L_e = kD/f$. Example: k = 5, D = 0.08 m, f = 0.04 → $L_e = 10$ m.

3. Pipes in series and in parallel (like electric circuits)

Arrangement	Discharge	Head loss	Electrical analogy
Series (one after another)	Same Q in every pipe	Losses **add**: $H = h_{f1} + h_{f2} + \dots$	Resistances in series
Parallel (branches between two junctions)	$Q = Q_1 + Q_2 + \dots$	**Same** loss in every branch	Resistances in parallel
Equivalent pipe (Dupuit)	Same Q, same total loss	$L/D^5 = L_1/D_1^5 + L_2/D_2^5 + \dots$	Single equivalent resistor

In parallel branches with the same f and L, $h_f \propto Q^2/D^5$, so $\mathbf{Q \propto D^{2.5}}$. A branch of twice the diameter carries $2^{2.5} \approx 5.66$ times the discharge (this is how 2023 Q88 was solved).

4. Power transmitted through a pipe

Power at the outlet $P = \rho\,g\,Q\,(H - h_f)$. It is **maximum when $h_f = H/3$**, and the maximum efficiency is $(H - H/3)/H =$ **66.7 %**. (2023 asked this twice.)

5. Water hammer

When a valve at the end of a long pipe is closed quickly, the moving column of water is stopped; its momentum turns into a sharp **pressure rise** that travels up and down the pipe as a wave (speed C ≈ 1400 m/s in water). It can burst the pipe. Surge tanks, slow-closing valves and relief valves protect against it.

Water hammer

- Time for the wave to go and return: 2L/C.
- **Gradual closure** (closing time t > 2L/C): p = ρ L V / t.
- **Sudden closure** (t < 2L/C), rigid pipe: p = ρ V C.
- 2023 Q93: L = 2500 m, V = 1.5 m/s, t = 25 s, C = 1460 m/s → 2L/C = 3.4 s < 25 s → gradual: p = 1000 × 2500 × 1.5/25 = 1.5 × 10^5 N/m² = 15 N/cm².

6. Confusing areas and exam traps

- Check whether the given f is Darcy's f or the friction coefficient f′ (= f/4).
- Sudden enlargement loss uses $(V_1 - V_2)^2$, not $V_1^2 - V_2^2$.
- In the water-hammer formula, first compare t with 2L/C; then pick the formula.
- In a horizontal pipe of constant diameter with only friction, the **velocity head and datum head stay constant** while pressure head falls (2023 Q86).

7. Connections

Pipe networks → water supply distribution design, Hardy Cross method (Paper 3 Civil environmental engineering). Water hammer → surge tanks in hydro plants (Topic 10).

Memory code: "One-third lost, two-thirds delivered"

1/3 lost	Maximum power when friction loss h_f = H/3; maximum efficiency = 2/3 = 66.7 %
½ in, 1 out	Entrance loss 0.5 V²/2g; exit loss 1.0 V²/2g
Series adds Loss, Parallel adds Q	Series → same Q, losses add · Parallel → same loss, Q adds
Slow = ρLV/t, Sudden = ρVC	Water hammer formulas

Self-recall (close the notes, answer aloud, then check)

1. Write the Darcy–Weisbach equation and f for laminar flow.
   $h_f = fLV^2/(2gD)$; $f = 64/Re$.
2. A 200 mm pipe suddenly enlarges to 400 mm; V_1 = 4 m/s. Loss of head (g = 10)?
   V_2 = 1 m/s; $h = (4 - 1)^2/20 = 0.45$ m.
3. What is the maximum efficiency of power transmission through a pipe?
   66.7 %, when $h_f = H/3$.

Quick check: 5 questions

1. A pipe of 200 mm diameter carrying water at 4 m/s is suddenly enlarged to 400 mm diameter. Taking g = 10 m/s², the loss of head due to the enlargement is

   1. 0.80 m
   2. 0.15 m
   3. 0.45 m
   4. 0.36 m

   1. Uses $V_1^2/2g$ — forgets to subtract V_2.
   2. Forgets to square: $(V_1 - V_2)/2g$ = 3/20.
   3. Correct. $V_2 = 4 \times (200/400)^2 = 1$ m/s; $h_e = (V_1 - V_2)^2/2g = 9/20 = 0.45$ m.
   4. Check V_2: with the diameter doubled, $V_2 = V_1/4$ = 1 m/s.

2. The diameter of a pipe is suddenly increased to three times. The ratio of the loss of head to the velocity head in the smaller pipe is

1. 64/81
2. 8/9
3. 1/9
4. 81/64

1. Correct. $V_2 = V_1/9 \rightarrow$ loss $= (1 - 1/9)^2 V_1^2/2g = (8/9)^2 = 64/81$ of the velocity head (2023 asked the same idea with a doubled diameter: 9/16).
2. Forgets to square.
3. This is V_2/V_1, not the loss ratio.
4. Inverted.

3. Consider the following statements about pipe flow:

I. For laminar flow in a pipe, the Darcy friction factor f = 64/Re.
II. For pipes in parallel, the head loss is the same in each branch.
III. For pipes in series, the total discharge is the sum of the discharges in each pipe.
IV. Power transmitted through a pipe is maximum when the friction loss is one-third of the total head.

Which of the above statements are correct?

1. I and II only
2. II, III and IV only
3. I, II, III and IV
4. I, II and IV only

1. Leaves out IV.
2. Includes III — in series the discharge is the same in every pipe.
3. Includes the false statement III.
4. Correct. I, II and IV are true; III is false.

4. Water flows at 2 m/s in a pipe 2 km long. A valve at the end is closed in 20 s. Taking the pressure-wave velocity as 1400 m/s, the rise in pressure is

1. 280 N/cm²
2. 20 N/cm²
3. 2 N/cm²
4. 10 N/cm²

1. This is the sudden-closure value ρVC — but 2L/C = 2.86 s is less than 20 s, so closure is gradual.
2. Correct. Gradual closure: $p = \rho LV/t = 1000 \times 2000 \times 2/20 = 2 \times 10^5$ N/m² = 20 N/cm².
3. Unit slip of 10.
4. Halves the correct value.

5. An oil of kinematic viscosity 0.2 stoke flows through a pipe of 5 cm diameter. Taking the critical Reynolds number as 2000, the critical velocity is

1. 0.4 m/s
2. 0.8 m/s
3. 1.6 m/s
4. 8 m/s

1. Uses a diameter of 10 cm.
2. Correct. $V = Re \cdot \nu/D = 2000 \times 0.2 \times 10^{-4} / 0.05 = 0.8$ m/s.
3. Uses a diameter of 2.5 cm.
4. Converts stoke wrongly (10^{-3} instead of 10^{-4}).

Topic 7: Boundary Layer, Drag and Lift

Syllabus line: *Boundary layer — thickness, laminar and turbulent boundary layers, separation; drag and lift.*

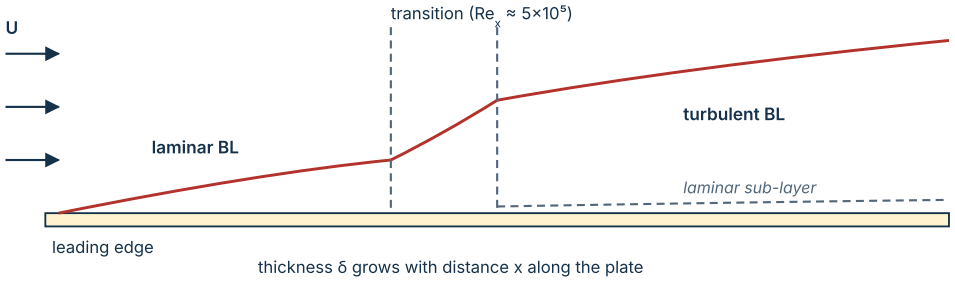
Asked in APPSC AEE 2023, Paper II (verified): 11 questions

- Length at which the boundary layer becomes turbulent (Q140); boundary layer thickness on a plate — 15.5 mm and 92.1 mm (Q145, Q146).
- Momentum thickness for a linear profile u/U = y/δ — δ/6 (Q141); displacement thickness after separation increases (Q149).
- Statement types: BL thickness vs Reynolds number (Q142), von Kármán momentum equation (Q144), turbulent flow (Q147), real fluid (Q148), laminar sub-layer and BL thickness definition (Q150).

- Resistance of a flat plate moving normal to water, C_D = 1.15 (Q143).

1. Basic idea

When fluid flows over a solid surface, the layer touching the surface sticks to it (no-slip, Topic 1). A little away from the surface the fluid moves at nearly the full stream velocity U. The thin region in between, where the velocity rises from 0 to 0.99U, is the **boundary layer** (Prandtl, 1904). Outside it, viscosity can be ignored; inside it, viscosity is everything.



The boundary layer starts laminar at the leading edge, thickens with distance, becomes turbulent beyond $Re_x \approx 5 \times 10^5$, and keeps a very thin laminar sub-layer next to the wall.

Essential formulas — boundary layer on a flat plate

- Local Reynolds number $Re_x = U\,x / \nu$; laminar up to about **5×10^5** → transition distance $x = 5\times10^5\,\nu / U$.
- **Laminar (Blasius)**: $\delta = 5x / \sqrt{Re_x}$ · **Turbulent**: $\delta = 0.37\,x / Re_x^{1/5}$
- δ grows along x; for a given x, δ/x **decreases** as Re increases (faster flow → thinner layer). So 'BL thickness increases with Re' is false (2023 Q142).
- **Displacement thickness** $\delta^* = \int(1 - u/U)dy$ · **momentum thickness** $\theta = \int(u/U)(1 - u/U)dy$ · **energy thickness** = $\int(u/U)(1 - u^2/U^2)dy$
- Linear profile u/U = y/δ: $\delta^* = \delta/2$, $\theta = \delta/6$, energy thickness = δ/4. Shape factor $H = \delta^*/\theta$ (= 3 here; Blasius ≈ 2.59).
- **Von Kármán momentum integral**: $\tau_0 / (\rho U^2) = d\theta/dx$ — gives skin-friction drag for laminar and turbulent layers.

2. Separation — why it matters

Where the outside flow slows down, pressure rises in the flow direction (**adverse pressure gradient**, dp/dx > 0). The slow fluid near the wall cannot push against it; it stops and turns back. At the **separation point** the wall velocity gradient $(du/dy)_{y=0} = 0$. Behind it forms a wide, eddying **wake** with low pressure — this causes large **pressure (form) drag**. Separation is delayed by streamlining, suction, slots, or making the layer turbulent (dimples on a golf ball). Displacement thickness **increases** after separation.

Drag and lift

- **Drag** (along the flow): $F_D = C_D\,A\,\rho U^2/2$ · **Lift** (normal to the flow): $F_L = C_L\,A\,\rho U^2/2$
- Total drag = **skin-friction drag** (shear on the surface) + **pressure (form) drag** (due to the wake). Streamlined body → mostly friction drag; bluff body → mostly pressure drag.
- 2023 Q143: plate 1 m × 1 m normal to water at 5 m/s, C_D = 1.15 → $F = 1.15 \times 1 \times 1000 \times 25/2 = 14{,}375$ N.
- Very slow flow past a sphere (Re < 1): **Stokes' law** $F_D = 3\pi\mu DV$ (settling of fine particles — sedimentation tanks).

3. Everyday analogy

Ride a bike: hold your palm flat to the wind and it is pushed back hard (bluff body, big wake → pressure drag). Turn the palm edge-on and the push almost disappears (streamlined → only skin friction). Tilt it slightly and it lifts up — that is lift.

4. Confusing areas and exam traps

- BL thickness is defined at u = 0.99U (not 0.19U — a 2023 distractor).
- The laminar sub-layer exists **inside the turbulent** boundary layer, not in a laminar one.
- Turbulent BL is thicker and has higher skin friction, but it resists separation better — so it often **reduces total drag** on bluff bodies.

5. Connections

Stokes' law → design of grit chambers and sedimentation tanks (Paper 3 Civil environmental engineering). Drag → wind loads on buildings and bridges. Lift → aircraft wings, turbine and pump blades (Topics 8–9).

Memory code: "D-E-M = 2-4-6" for a linear profile

D = 2	Displacement thickness $\delta^* = \delta/2$
E = 4	Energy thickness = $\delta/4$
M = 6	Momentum thickness $\theta = \delta/6$
5 lakh	Transition on a flat plate at $Re_x \approx 5 \times 10^5$
Zero slope = Separation	At the separation point $(du/dy)_{wall} = 0$

Self-recall (close the notes, answer aloud, then check)

1. Define displacement thickness in words.
   The distance by which the wall would have to be moved outward so that an ideal (no-friction) flow carries the same discharge as the real flow with its velocity deficit.
2. When does a boundary layer separate?
   Under an adverse pressure gradient ($dp/dx > 0$) when the wall shear becomes zero, $(du/dy)_{y=0} = 0$.
3. Name the two parts of total drag.
   Skin-friction (shear) drag and pressure (form) drag.

Quick check: 5 questions

1. For a linear velocity profile $u/U = y/\delta$ in a boundary layer, the ratio of displacement thickness to momentum thickness is

   1. 2 : 1
   2. 3 : 1
   3. 6 : 1
   4. 3 : 2

   1. This is $\delta/2 \div \delta/4$ — it uses the energy thickness instead of the momentum thickness.
   2. Correct. $\delta^* = \delta/2$ and $\theta = \delta/6$, so $\delta^*/\theta = 3$ (the shape factor).
   3. Uses $\delta \div \theta$.
   4. This is $\delta/4 \div \delta/6$ — energy thickness to momentum thickness.

2. Air ($\nu = 1.5 \times 10^{-5}$ m²/s) flows at 15 m/s over a smooth flat plate. Taking the transition Reynolds number as 5×10^5, the boundary layer becomes turbulent at a distance of about

   1. 0.05 m
   2. 0.25 m
   3. 5 m
   4. 0.5 m

   1. Ten times too small — check the powers of ten.
   2. Uses $Re = 2.5 \times 10^5$.
   3. Ten times too large.
   4. Correct. $x = Re \cdot \nu/U = 5 \times 10^5 \times 1.5 \times 10^{-5} / 15 = 0.5$ m.

3. Consider the following statements about the boundary layer:
   I. Separation occurs under an adverse pressure gradient.
   II. At the point of separation, the velocity gradient at the wall is zero.
   III. The boundary layer thickness decreases along the length of a flat plate.
   IV. A laminar sub-layer exists within a turbulent boundary layer.

   Which of the above statements are correct?

   1. I, II and IV only
   2. I, II and III only
   3. I and II only
   4. I, II, III and IV

   1. Correct. I, II and IV are true; III is false — δ grows with distance x.
   2. Includes III, which is false.
   3. Leaves out IV.
   4. Includes III.

4. A plate 1 m × 0.5 m is held normal to a stream of water flowing at 2 m/s. If the drag coefficient is 1.2, the drag force is

1. 600 N
2. 1200 N
3. 2400 N
4. 4800 N

1. Uses $\rho U^2/4$.
2. Correct. $F = C_D A \rho U^2/2 = 1.2 \times 0.5 \times 1000 \times 4/2 = 1200$ N.
3. Forgets the ½.
4. Uses U = 4 m/s.

5. **Assertion (A):** The dimples on a golf ball help it travel farther.
Reason (R): Dimples make the boundary layer turbulent, which delays separation and reduces the pressure drag.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. Both are true, and R explains A: a smaller wake means less pressure drag.
2. R is the actual reason, not an unrelated fact.
3. R is true: turbulent layers carry more momentum near the wall and resist separation.
4. A is true.

Topic 8: Impact of Jets and Hydraulic Turbines

Syllabus line: *Impact of jets on vanes, velocity diagrams; turbines — Pelton, Francis, Kaplan, draft tube; performance — unit and specific quantities, characteristic curves, governing, cavitation.*

Asked in APPSC AEE 2023, Paper II (verified): 8 questions

- Match turbines with specific speeds — Pelton (one jet) 10–35, Pelton (two jets) 35–60, Francis 60–300, Kaplan 300–1000 (Q121).
- Power developed when moment of momentum falls by 12 kN·m at 600 rpm — 754 kW (Q123).
- Statements: pressure in impulse vs reaction turbines (Q124); match nozzle size, pressure, velocity, cavitation erosion (Q125).
- Work done per second by a jet on a series of flat plates (Q126); force of an oil jet on a stationary plate (Q127); force of a nozzle jet with $C_V = 0.95$ (Q128); match bucket, guide blade, volute, percent slip (Q139).

1. Basic idea

A jet of water has momentum. When it hits a plate or vane and changes direction, the plate pushes on the water and the water pushes back on the plate (Newton's third law). F = ρQ × (change of velocity). A **turbine** is simply many vanes on a wheel, arranged so that this force keeps turning a shaft.

Essential formulas — impact of jets (a = jet area, V = jet velocity, u = plate velocity)

- Stationary flat plate, normal: $F = \rho a V^2$ · inclined at θ to the jet: $F_n = \rho a V^2 \sin\theta$
- Single moving flat plate: $F = \rho a (V - u)^2$ (the plate runs away from part of the water)
- **Series of flat plates** on a wheel: $F = \rho a V (V - u)$; work/s = $\rho a V (V - u) u$; **maximum efficiency 50 % at u = V/2**
- Series of curved (semicircular) vanes: maximum efficiency 100 % in theory, at u = V/2 — why Pelton buckets are curved
- Moment of momentum → torque T; power $P = 2\pi NT/60$. 2023 Q123: $2\pi \times 600 \times 12/60 = 754$ kW.

2. Types of turbines

Feature	Pelton wheel	Francis turbine	Kaplan turbine
Type	**Impulse** — all head turned into a jet in the nozzle; runner at atmospheric pressure	**Reaction** — pressure falls inside the runner; runs full	**Reaction**, axial flow
Head	High	Medium	Low
Discharge	Small	Medium	Large

Feature	Pelton wheel	Francis turbine	Kaplan turbine
Flow direction	Tangential	Radial inward / mixed	Axial
Specific speed N_s (SI)	10–35 (one jet), 35–60 (more jets)	60–300	300–1000
Governing	Spear (needle) valve + jet deflector	Guide vanes (wicket gates)	Guide vanes + adjustable runner blades (double regulation)
Draft tube	Not used	Needed	Needed

Performance — unit and specific quantities

- **Specific speed** (turbine) $N_s = N\sqrt{P} / H^{5/4}$ (P in kW) — the speed of a geometrically similar turbine producing 1 kW under 1 m head. It decides the turbine type.
- **Unit quantities** (under 1 m head): $N_u = N/\sqrt{H} \cdot Q_u = Q/\sqrt{H} \cdot P_u = P/H^{3/2}$ — used to predict performance at a different head.
- Efficiencies: hydraulic η_h · mechanical η_m · overall $\eta_o = \eta_h \times \eta_m$; shaft power $P = \eta_o \rho g Q H$.
- **Characteristic curves**: main (constant head), operating (constant speed — the normal running condition), constant-efficiency (muschel) curves.

3. Draft tube and cavitation

The **draft tube** is a gradually widening pipe from the runner exit to the tail race. It (1) lets the turbine be placed **above** the tail-water level for easy maintenance without losing that head, and (2) **recovers kinetic energy** of the exit water by slowing it down. Because it creates suction below the runner, the pressure can drop to vapour pressure → **cavitation** (pitting and noise). It is checked by Thoma's cavitation factor σ.

4. Confusing areas and exam traps

- Higher head → **lower** specific speed (Pelton). Low head, large flow → high N_s (Kaplan).
- Series of flat plates: 50 %. Series of curved vanes: up to 100 %. Both at u = V/2.
- Guide blades belong to Francis; buckets to Pelton; volute casing to centrifugal pumps; percent slip to reciprocating pumps (2023 Q139).

5. Connections

Hydro power layouts (Topic 10); pumps are reversed turbines (Topic 9); Paper 3 Mechanical — turbomachines; Paper 3 Electrical — hydroelectric stations.

Memory code: "Pin, Fan, Kite" — turbines climb in specific speed

Pin = Pelton	Smallest N_s (10–60) · highest head · impulse
Fan = Francis	Medium N_s (60–300) · medium head · radial/mixed reaction
Kite = Kaplan	Highest N_s (300–1000) · lowest head · axial, adjustable blades
Half speed, half/full efficiency	u = V/2 for maximum efficiency: flat plates 50 %, curved vanes 100 %

Self-recall (close the notes, answer aloud, then check)

1. Write the specific-speed formula for a turbine.
   $N_s = N\sqrt{P} / H^{5/4}$.
2. Give two functions of a draft tube.
   Allows the turbine to be set above tail water without loss of head; recovers kinetic energy at the runner exit.
3. How is a Pelton wheel governed?
   By moving the spear (needle) in the nozzle to change the jet area; a deflector diverts the jet during sudden load drops.

Quick check: 5 questions

1. A water jet of area 0.01 m² and velocity 20 m/s strikes a single flat plate held normal to it. The plate moves away from the jet at 5 m/s. The force on the plate is

1. 4000 N
2. 3000 N
3. 2250 N
4. 1500 N

1. This is the force on a stationary plate.
2. This is the force for a series of plates, ρaV(V − u).
3. Correct. Single moving plate: F = ρa(V − u)² = 1000 × 0.01 × 15² = 2250 N.
4. Halves the series-of-plates value; not a valid formula.

2. Match List-I with List-II:

List-I	List-II
A. Pelton wheel	1. Recovers exit kinetic energy; turbine can sit above tail water
B. Francis turbine	2. Axial flow with adjustable runner blades
C. Kaplan turbine	3. Tangential flow, impulse
D. Draft tube	4. Mixed (radial inward) flow, reaction

1. A-3, B-4, C-2, D-1
2. A-4, B-3, C-2, D-1
3. A-3, B-4, C-1, D-2
4. A-3, B-2, C-4, D-1

1. Correct. Pelton → tangential impulse; Francis → mixed-flow reaction; Kaplan → axial with adjustable blades; draft tube → recovers exit kinetic energy and lets the turbine sit above tail water.
2. Swaps Pelton and Francis.
3. Swaps Kaplan and the draft tube.
4. Swaps Francis and Kaplan.

3. A turbine runs at 300 rpm and develops 8100 kW under a head of 81 m. Its specific speed and the most suitable type are

1. 37, Pelton wheel
2. 111, Francis turbine
3. 111, Pelton wheel
4. 333, Kaplan turbine

1. Uses H instead of H to the power 5/4 with the wrong power term — recheck: √8100 = 90, 81^(5/4) = 243.
2. Correct. $N_s = N\sqrt{P}/H^{5/4} = 300 \times 90 / 243 = 111$ → Francis range (60–300).
3. Right number, wrong turbine — Pelton is below 60.
4. Uses $H^{3/4}$ instead of $H^{5/4}$.

4. Consider the following statements about hydraulic turbines:
I. In an impulse turbine, the water in the runner is at atmospheric pressure.
II. Reaction turbines run full and need a draft tube.
III. A Pelton wheel is governed by changing the angle of the guide vanes.
IV. A Kaplan turbine suits low heads and large discharges.
Which of the above statements are correct?

1. I, II, III and IV
2. I and II only
3. I, II and IV only
4. II, III and IV only

1. Includes III — Pelton is governed by a spear valve.
2. Leaves out IV.
3. Correct. I, II and IV are true; III is false.
4. Includes the false statement III.

5. For a jet striking a series of flat radial plates mounted on a wheel, the maximum efficiency and the condition for it are

1. 66.7 %, when u = V/3
2. 100 %, when u = V/2
3. 50 %, when u = V/3
4. 50 %, when u = V/2

1. Mixes up the pipe power-transmission result (66.7 %).
2. This is for curved (semicircular) vanes, not flat plates.
3. Right efficiency, wrong speed ratio.
4. Correct. Work/s = ρaV(V − u)u is maximum at u = V/2, giving η = 2u(V − u)/V² = 50 %.

Topic 9: Centrifugal Pumps and Hydraulic Control Valves

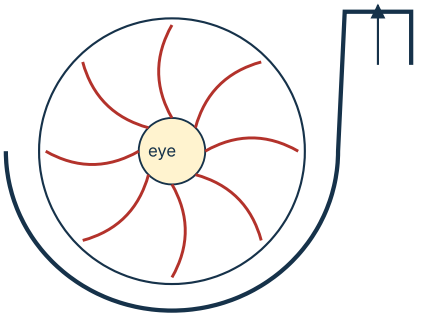
Syllabus line: *Centrifugal pumps — work done, efficiencies, specific speed, NPSH; hydraulic control valves.*

Asked in APPSC AEE 2023, Paper II (verified): 10 questions

- Power to drive a pump delivering 50 L/s against 40 m (Q129); a pump transfers energy from rotor to fluid (Q130).
- Number of identical pumps needed for a deep well (Q131, Q136) — from specific speed.
- Net head with radial outlet vanes, D = 100 mm, 3000 rpm, η_{mano} = 0.8 — 20 m (Q132); manometric efficiency (Q133).
- Discharge and head when speed is halved (Q134); head with a different liquid (Q135); series/parallel statements (Q137); rotodynamic pump statements (Q138).

1. Basic idea

A **centrifugal pump** is a turbine in reverse: a motor spins an **impeller**; its curved vanes fling water outward. The water gains velocity and pressure (energy from **rotor to fluid**), is collected in a spiral **volute casing** that slowly widens to turn velocity into pressure, and leaves through the delivery pipe.



Energy given to water per kg = $V_{w2}\ u_2$
(radial entry, no whirl at inlet)

Manometric efficiency $\eta_{mano} = g\ H_m / (V_{w2}\ u_2)$

$u_2 = \pi\ D_2\ N / 60$ (impeller tip speed)

Radial vanes at outlet: $V_{w2} = u_2$

Water enters the eye axially, is thrown outward by the vanes, and the widening volute converts its velocity head into pressure head.

Essential formulas

- Power required $P = \rho\ g\ Q\ H_m / \eta_o$. Example (2023 Q129): 1000 × 9.81 × 0.05 × 40 ≈ 19.6 kW (output power).
- Radial vanes at outlet: $H_m = \eta_{mano} \times u_2^2/g$. 2023 Q132: $u_2 = \pi \times 0.1 \times 3000/60$ = 15.7 m/s → 0.8 × 15.7²/9.81 ≈ 20 m.
- **Specific speed** (pump) $N_s = N\ \sqrt{Q}\ /\ H_m^{3/4}$ (Q in m³/s). Number of pumps: in series if the total head is too large for one pump; in parallel if the discharge is too large.
- **Affinity laws** (same pump, speed changed): $Q \propto N \cdot H \propto N^2 \cdot P \propto N^3$. Halve the speed → half the discharge, one-quarter the head, one-eighth the power.
- Minimum starting speed: $u_2^2/2g - u_1^2/2g = H_m$ (the centrifugal head must at least equal the manometric head).
- **NPSH** (net positive suction head) available = head at suction above vapour pressure. Keep $NPSH_{available} > NPSH_{required}$, else **cavitation**.

2. Practical points

- **Priming**: a centrifugal pump must be filled with liquid before starting — an impeller spinning in air cannot create enough suction (air is ~800 times lighter). A reciprocating pump is self-priming.
- **Series** pumps → heads add (multistage pumps for high head, e.g. deep borewells). **Parallel** pumps → discharges add.
- **Reciprocating pump**: positive displacement (piston). Slip = $Q_{theoretical} - Q_{actual}$; percent slip; negative slip possible with long delivery pipes; **air vessels** smooth the flow and save work.

Hydraulic control valve	Purpose
Gate (sluice) valve	Full-bore on/off isolation; low loss when fully open; not for throttling
Globe valve	Throttling / regulating flow; higher head loss
Non-return (check, reflux) valve	Allows flow in one direction only — stops back-flow when a pump stops
Butterfly valve	Quick quarter-turn shut-off in large pipes; compact
Pressure-relief / safety valve	Opens automatically to release excess pressure (water hammer protection)
Air-release valve	Lets trapped air escape at high points of pipelines
Foot valve with strainer	At the bottom of the suction pipe: keeps the pump primed and stops debris

3. Confusing areas and exam traps

- Pumps in parallel boost **discharge**; in series boost **head** (2023 Q137 — only statement 1 correct).
- Turbine N_s uses $\sqrt{P}$ and $H^{5/4}$; pump N_s uses $\sqrt{Q}$ and $H^{3/4}$.
- Priming is needed for the centrifugal pump, not the reciprocating pump.

4. Connections

Water supply pumping stations (Paper 3 Civil — environmental engineering); lift irrigation schemes; Paper 3 Mechanical — turbomachines.

Memory code: "Q-H-P = 1-2-3"

$Q \propto N^1$	Discharge varies with speed
$H \propto N^2$	Head varies with square of speed
$P \propto N^3$	Power varies with cube of speed
Series = Stack heads	Pumps in series add head; parallel add discharge

Self-recall (close the notes, answer aloud, then check)

1. Why is priming necessary for a centrifugal pump?
   The head developed depends on the density of the fluid in the impeller; with air inside, the suction created is far too small to lift water.
2. A pump's speed is doubled. What happens to Q, H and P?
   Q doubles, H becomes 4 times, P becomes 8 times.
3. What is NPSH?
   The net head available at the pump suction above the vapour pressure of the liquid; it must exceed the NPSH required to avoid cavitation.

Quick check: 5 questions

1. A centrifugal pump delivers 30 L/s against a head of 20 m at 1500 rpm. If the speed is raised to 3000 rpm, the discharge and head become
   1. 60 L/s and 40 m
   2. 120 L/s and 80 m
   3. 60 L/s and 80 m
   4. 30 L/s and 80 m

   1. Treats the head as varying with N — it varies with N^2.
   2. Squares the discharge ratio as well.
   3. Correct. $Q \propto N$ → 60 L/s; $H \propto N^2$ → 20 × 4 = 80 m.
   4. Discharge also changes with speed.

2. A pump delivers 0.05 m^3/s of water against a manometric head of 30 m. If its overall efficiency is 75 % (g = 10 m/s^2), the power required to drive it is

1. 15 kW
2. 11.25 kW
3. 26.7 kW
4. 20 kW

1. This is the power given to the water (output), not the input.
2. Multiplies by the efficiency instead of dividing.
3. Uses a head of 40 m (0.05 × 40 × 10 / 0.75 = 26.7).
4. Correct. P = ρgQH/η = 1000 × 10 × 0.05 × 30/0.75 = 20,000 W = 20 kW.

3. Consider the following statements about pumps:
I. Pumps connected in series increase the head.
II. Pumps connected in parallel increase the discharge.
III. A reciprocating pump must be primed before starting.
IV. Cavitation is avoided by keeping the available NPSH greater than the required NPSH.

Which of the above statements are correct?

1. I, II, III and IV
2. I, II and IV only
3. I and II only
4. II, III and IV only

1. Includes III — priming is for centrifugal pumps.
2. Correct. I, II and IV are true; III is false.
3. Leaves out IV.
4. Includes III.

4. Match List-I with List-II:

List-I	List-II
A. Centrifugal pump	1. Air vessel
B. Reciprocating pump	2. Volute casing
C. Non-return (check) valve	3. Full-bore on/off isolation
D. Gate (sluice) valve	4. Prevents back-flow

1. A-2, B-1, C-4, D-3
2. A-1, B-2, C-4, D-3
3. A-2, B-1, C-3, D-4
4. A-1, B-2, C-3, D-4

1. Correct. Centrifugal pump → volute casing; reciprocating pump → air vessel; non-return valve → prevents back-flow; gate valve → full-bore isolation.
2. Swaps the two pumps.
3. Swaps the two valves.
4. Swaps both pairs.

5. An impeller of 0.2 m outer diameter runs at 1500 rpm and has radial vanes at the outlet. If the manometric efficiency is 0.8 (g = 10 m/s^2), the manometric head developed is about

1. 10 m
2. 20 m
3. 25 m
4. 31 m

1. Halves the head — uses $u_2^2/2g$.
2. Correct. $u_2 = \pi \times 0.2 \times 1500/60 = 15.7$ m/s; $H_m = 0.8 \times u_2^2/g = 0.8 \times 24.7 \approx 20$ m.
3. Ignores the manometric efficiency.
4. Would need a higher tip speed; recheck $u_2 = \pi DN/60$.

Topic 10: Hydro-Power Projects — Layouts, Storage, Dams and Spillways

Syllabus line: *Hydro projects and power plants — layouts, pumped storage, hydrographs, storage and pondage, dams and spillways.*

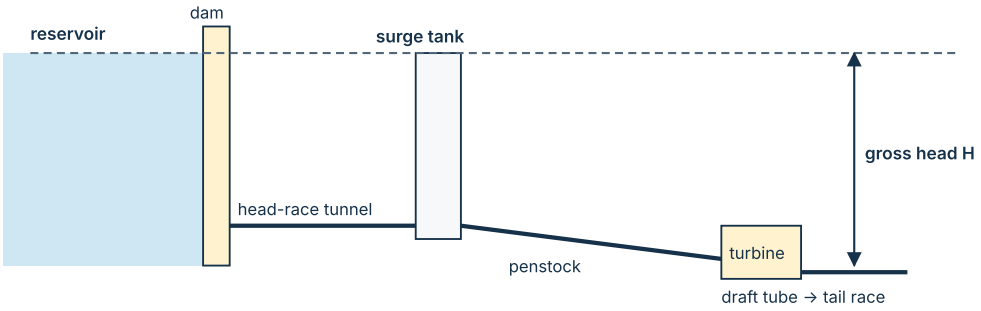
Asked in APPSC AEE 2023, Paper II (verified): 1 question

- Correct sequence of installations in the direction of flow in a hydropower plant (Q119).

Our analysis:
only one question came from this area in 2023, but it is a full syllabus line, links to Paper 3 Civil (hydrology, dams) and Paper 3 Electrical (generating stations). Learn the layout, the role of each part and the definitions below.

1. Basic idea

A hydro-power plant turns the potential energy of stored water into electricity: P = η ρ g Q H. You need a **head** (height difference, created by a dam or a natural fall) and a **discharge** (river flow, regulated by storage).



Direction of flow: reservoir → intake → head-race tunnel → surge tank → penstock → turbine → draft tube → tail race. (The 2023 key accepted 'reservoir, penstock, surge tank, turbine' — in that arrangement the surge tank stands on the penstock near the power house. Pick the option that keeps reservoir first and turbine last with the surge tank before the turbine.)

Part	Job
Dam / weir	Creates head and storage
Intake + trash rack	Admits water, keeps out floating debris
Head-race tunnel / canal	Carries water at low pressure to the surge tank or forebay
Surge tank	Absorbs water-hammer pressure rise on sudden load rejection; supplies extra water on sudden load increase
Penstock	Steel pipe carrying water at high pressure to the turbine
Turbine + generator	Converts hydraulic energy to mechanical, then electrical energy
Draft tube + tail race	Recover exit energy and return water to the river

2. Classification of hydro plants

- **Run-of-river** (no or little storage) — uses the river flow as it comes; **with pondage** it can store water for a few hours or days behind a small weir to meet **daily peaks**.
- **Storage** plant — a large reservoir stores monsoon water for use through the dry months.
- **Pumped storage** — two reservoirs at different levels. In off-peak hours surplus power pumps water up; in peak hours the water generates power. It **consumes more energy than it gives back** (about 70–80 % is recovered) but turns cheap off-peak power into valuable peak power. Reversible pump-turbines are common.
- By head (commonly quoted): low head (< 30 m, Kaplan/propeller), medium head (30–300 m, Francis), high head (> 300 m, Pelton). Books differ slightly on the limits.

3. Hydrographs, storage and pondage

- **Hydrograph**: graph of river discharge against time. Its area = volume of water.
- **Flow-duration curve**: discharge against the percentage of time it is equalled or exceeded — tells the firm (dependable) power.
- **Mass curve** (Rippl's method): cumulative inflow against time; tangents drawn parallel to the demand line give the **storage capacity** needed.
- **Storage** = long-term (seasonal) holding of water in a reservoir; **pondage** = short-term (hours to a week) holding to follow load changes.

4. Dams and spillways

- **Dams**: gravity (resists water by its own weight — masonry/concrete), arch (transfers load to the valley sides — narrow rocky gorges), buttress, earth and rock-fill (wide valleys, weak foundations).

- **Spillway**: a safety valve that passes surplus flood water so it never overtops the dam. Types: ogee (overflow), chute, side-channel, shaft (morning-glory), siphon. Energy at the toe is destroyed by stilling basins (hydraulic jump), buckets.

5. Connections

Paper 3 Civil — hydrology (hydrographs, reservoir capacity), irrigation (gravity dams, spillways, energy dissipaters). Paper 3 Electrical — generating stations, load curves and pumped storage.

Memory code: "R-I-H · S-P · T-D-T" — say it as three beats, following the water downhill

R-I-H	Reservoir → Intake → Head-race
S-P	Surge tank → Penstock
T-D-T	Turbine → Draft tube → Tail race
Pondage = Pocket money	Short-term (daily/weekly) storage; storage = savings account (seasonal)

Self-recall (close the notes, answer aloud, then check)

1. Give two functions of a surge tank.
   Relieves water-hammer pressure on sudden load drop; supplies water quickly on sudden load increase.
2. Is a pumped-storage plant a net producer of energy?
   No — it consumes more energy in pumping than it generates; it shifts energy from off-peak to peak hours.
3. Which curve gives the reservoir capacity needed for a given demand?
   The mass curve (Rippl's method).

Quick check: 5 questions

1. The correct order of components in the direction of flow in a hydro-electric scheme with a long tunnel is

   1. Reservoir → penstock → head-race tunnel → surge tank → turbine → draft tube
   2. Reservoir → head-race tunnel → surge tank → penstock → turbine → draft tube
   3. Reservoir → surge tank → head-race tunnel → turbine → penstock → draft tube
   4. Reservoir → head-race tunnel → penstock → draft tube → turbine → surge tank

   1. The penstock comes after the tunnel, just before the turbine.
   2. Correct. Low-pressure tunnel first, surge tank where it meets the steep high-pressure penstock, then turbine and draft tube.
   3. The penstock always ends at the turbine.
   4. The draft tube is after the turbine and the surge tank is upstream.

2. Consider the following statements about pumped-storage plants:
   I. They pump water to the upper reservoir using surplus power in off-peak hours.
   II. They generate power during peak-demand hours.
   III. They produce more energy than they consume over a cycle.
   IV. Reversible pump-turbines are commonly used in them.

   Which of the above statements are correct?

   1. I, II, III and IV
   2. I and II only
   3. II, III and IV only
   4. I, II and IV only

   1. Includes III — they are net consumers.
   2. Leaves out IV.
   3. Includes the false statement III.
   4. Correct. I, II and IV are true; III is false.

3. Match List-I with List-II:

List-I	List-II
A. Surge tank	1. Finds the reservoir storage capacity
B. Draft tube	2. Passes surplus flood water safely
C. Spillway	3. Recovers kinetic energy at the runner exit
D. Mass curve	4. Relieves water-hammer pressure

1. A-4, B-3, C-1, D-2
2. A-3, B-4, C-1, D-2
3. A-4, B-3, C-2, D-1
4. A-3, B-4, C-2, D-1

1. Swaps spillway and mass curve.
2. Swaps surge tank and draft tube, and spillway and mass curve.
3. Correct. Surge tank → water hammer relief; draft tube → recovers exit kinetic energy; spillway → passes surplus flood; mass curve → storage capacity.
4. Swaps surge tank and draft tube.

4. **Assertion (A):** A run-of-river plant with pondage can meet the daily peak load.
Reason (R): Pondage stores water behind a small weir or barrage for a few hours to a few days.

1. Both A and R are true, and R is the correct explanation of A
2. Both A and R are true, but R is NOT the correct explanation of A
3. A is true, but R is false
4. A is false, but R is true

1. Correct. Both are true, and R explains A: water stored during low-load hours is released during peak hours.
2. R directly explains A.
3. R is the definition of pondage.
4. A is true.

5. A site has a head of about 400 m and a small discharge. The most suitable turbine is

1. Kaplan turbine
2. Propeller turbine
3. Francis turbine
4. Pelton wheel

1. Kaplan suits low heads and large discharges.
2. Propeller (fixed-blade Kaplan) is also a low-head turbine.
3. Francis is for medium heads.
4. Correct. Very high head with small discharge → impulse turbine (Pelton), lowest specific speed.

Notebook page — write these from memory

Close this page. In your notebook, write the items below in this order without looking. Then open the page and correct yourself in a different colour. Whatever you could not write is your revision list for Sunday.

Topic 1

1. Table of properties with units: ρ, w, S, μ, ν, σ, vapour pressure, K
2. τ = μ du/dy and the plate diagram; poise and stoke conversions
3. Viscosity vs temperature for liquids and gases
4. 4σ/d, 8σ/d, 2σ/d and capillary rise h = 4σcosθ/ρgd

Topic 2

1. Pascal's law, hydrostatic law p = ρgh
2. Absolute–gauge–vacuum diagram with 101.325 kN/m² = 10.3 m water = 760 mm Hg
3. Manometer rule (down add, up subtract) and the 12.6x result
4. List of devices and what each can and cannot measure

Topic 3

1. $F = \rho g A \bar{x}$ and $h^* = \bar{x} + I_G \sin^2\theta/(A\bar{x})$; pressure triangle with 2H/3
2. Curved surface: F_H and F_V rules
3. Archimedes' principle; depth of immersion
4. GM = I/V − BG; stability conditions for floating and submerged bodies; $T = 2\pi\sqrt{(k^2/g \cdot GM)}$

Topic 4

1. Six pairs of flow types with one example each
2. Continuity: AV = constant and $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$
3. Local and convective acceleration
4. Rotation, vorticity (= 2ω), irrotational condition
5. ψ and φ table: definitions, when each exists; flow net and its uses
6. Streamline, pathline, streakline

Topic 5

1. Bernoulli's equation, three heads, assumptions (S-I-N-S)
2. Sketch of venturimeter with TEL and HGL
3. Formulas and C_d values: venturimeter, orifice meter, pitot tube
4. Rotameter and hot-wire anemometer in one line each
5. Momentum and moment-of-momentum equations; α = 2, β = 4/3
6. Forced and free vortex; $z = \omega^2 R^2/2g$

Topic 6

1. Reynolds number limits; laminar and turbulent velocity profiles
2. Darcy–Weisbach, f = 64/Re, Hagen–Poiseuille, Chezy
3. All minor-loss formulas; equivalent length
4. Series, parallel and equivalent pipe rules
5. Maximum power: $h_f = H/3$, η = 66.7 %
6. Water hammer: 2L/C test, ρLV/t and ρVC

Topic 7

1. Sketch of boundary layer growth on a flat plate with $Re_x = 5 \times 10^5$
2. δ formulas (laminar, turbulent); δ*, θ, energy thickness and linear-profile values
3. Separation: cause, condition, control
4. Drag and lift formulas; friction vs pressure drag; Stokes' law

Topic 8

1. Jet force formulas: stationary, inclined, moving, series of plates
2. Comparison table Pelton / Francis / Kaplan with specific speed ranges
3. $N_s = N\sqrt{P}/H^{5/4}$ and unit quantities
4. Draft tube functions; cavitation; governing of each turbine

Topic 9

1. Parts of a centrifugal pump and energy conversion
2. $\eta_{mano} = gH_m/(V_{w2}u_2)$; $N_s = N\sqrt{Q}/H^{3/4}$
3. Affinity laws Q-H-P = 1-2-3; series and parallel
4. Priming, NPSH and cavitation; table of valves

Topic 10

1. Layout in flow order (R-I-H · S-P · T-D-T) with the job of each part
2. Run-of-river, storage, pumped storage
3. Hydrograph, flow-duration curve, mass curve; storage vs pondage
4. Types of dams and spillways

References

- APPSC Notification No. 27/2026 (Assistant Executive Engineers) — scheme and syllabus of Paper 2.
- APPSC AEE Paper II (Civil & Mechanical Engineering), held on 22 Aug 2023, Shift 1 — official paper with key. All "Asked in APPSC AEE 2023" notes refer to this paper.
- R.K. Bansal, *A Textbook of Fluid Mechanics and Hydraulic Machines* — sign conventions and formula forms follow this book, as the 2023 key did.
- Photos: Wikimedia Commons, free licences; each photo links to its source page.
- Quiz questions are original practice questions (Sulabham), not previous-year questions.

Facts verified on 10 Nov 2026.

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